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Title: Varicella - Zoster Immunoglobulin (VZIG) in the intramuscular (IM) form in Royal Hospital for children weighing 2.0 Kg

Introduction:

The Varicella - zoster virus is a highly contagious virus with airborne transmission capability. The infectious period is 1 to 2 days before the onset of the rash and ends when all lesions are crusted, typically 4 to 7 days after onset of rash in immunocompetent people, but this period may be longer in immunocompromised people. The infection can be severe and result in complications like; pneumonitis, encephalitis, DIC, necrotizing fasciitis and even death especially in immunocompromised population. Varicella vaccine is life attenuated and so can't be used for children who are immunocompromised and this group is left with the option of varicella specific immunoglobulin after reporting exposure to infection.

Information about VZIG:

VZIG is indicated for prevention of Chicken pox.It reduces the severity of chicken pox (varicella zoster virus) infections in high risk individuals after exposure. It should be given as soon as possible after VZV exposure (ideally within 96 hours) and may be given within 10 days of exposure. The Second dose can be considered in high-risk individuals if another VZV exposure occurs ≥3 weeks after the first dose.

Indications for VIZIG:

The VZIG generally is recommended after exposure to varicella infection for:

- Newborn infants whose mothers have signs and symptoms of varicella around the time of delivery (i.e., 5 days before to 2 days after).
- Hospitalized premature infants born at ≥28 weeks of gestation whose mothers do not have evidence of immunity to varicella.
- Hospitalized premature infants born at <28 weeks of gestation or who weigh ≤1,000 g at birth, regardless of their mothers' evidence of immunity to varicella.
- Pregnant women without evidence of immunity
- HIV patients
- Primary and secondary immunocompromised patients who are non-immune to varicella

Procedure for giving IM VZIG:

- Do *not* mix with any other drug or solution.
- Depending on patient weight, divide the IM dose and administer in 2 or more injection sites; not to exceed 3 ml/injection site.

Table 1. IM Dosage of VZIG (Varizig) for Neonates, Infants, Children, and Adolescents Based on Weight

Weight	Dosage	Number of Reconstituted Vials (125 Units)
2.1–10 kg	125 units	1 vial (1.2 ml)
10.1–20 kg	250 units	2 vials (2.4 ml)
20.1–30 kg	375 units	3 vials (3.6 ml)
30.1–40 kg	500 units	4 vials (4.8 ml)
≥40.1 kg	625 units	5 vials (6 ml)

- The patient has to be isolated in a single room with airborne precaution.
- It can be given by staff nurse in OPD, ER or other ward till receive Immunoglobulin and discharge home as soon as possible.
- Hospitalized children exposed to varicella and given VZIG should be kept in strict isolation for 28 days after exposure since the prophylaxis is not 100% efficacy in disease prevention.

Contraindications:

- Hypersensitivity to immune globulin product.
- Severe thrombocytopenia (IM injection).
- Known immunity to varicella zoster virus.

Precautions for use:

IgA deficient individual (there is a potential risk of hypersensitivity reactions).

Adverse effects:

- Local discomfort, redness and swelling at the injection site (1%).
- Headache (1%).
- Rash (1%).
- Rare adverse reaction Myalgia, rigor, fatigue, nausea and flushing (0.2%)

Drug interaction:

Interferes with immune response to live vaccine, such as measles, mumps and rubella, defer administration of live vaccines more than 6 months after VZIG dose.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Food and Drug Administration. FDA approves VariZIG for reducing chickenpox symptoms. Available at http://www.fda.gov/newsevents/newsroom/pressannouncements/ucm333233.htm .	Silver Spring	2012	

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American Academy of Pediatrics. Varicella-zoster infections. I. Red book: 2012 report of the Committee on Infectious Diseases.	Pickering LK,Baker CJ..etc	2012	774-89	

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